

A-level

# ENVIRONMENTAL SCIENCE

## Questions by topic: Conservation priorities, legislation & protocols - Set 1

### Mark scheme

AQA spec 3.1.2.3.1-3.1.2.3.2; The living environment; textbook pp. 26-39

### Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

**Textbook references:** every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Marks |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 01.1     | <ul style="list-style-type: none"> <li>• Coordinating global data on biodiversity conservation (Red List)</li> <li>• Increasing understanding of the importance of biodiversity</li> <li>• Deploying nature-based solutions to global challenges in climate, food and sustainable development</li> </ul>                                                                                                                                                                   | 3     |
| 01.2     | <ul style="list-style-type: none"> <li>• Decline = <math>(12\,000 - 2100) / 12\,000 \times 100 = 82.5\%</math></li> <li>• Decline over 80% meets the Critically Endangered criterion even though habitat area (400 km<sup>2</sup>) and adults (2100) only meet Endangered</li> <li>• A species is placed in the highest category for which it meets any one criterion, so Critically Endangered</li> </ul>                                                                 | 3     |
| 01.3     | <ul style="list-style-type: none"> <li>• Categorising a species requires a great deal of research (population, range, trends) and resources; about 85 000 of millions of species assessed</li> <li>• Key species in habitats under particular threat (eg lemurs indicating the state of Madagascar's forests)</li> <li>• Evolutionarily unique species / EDGE species; endemic species; keystone species; flagship species; species with fragmented populations</li> </ul> | 3     |
| 01.4     | <ul style="list-style-type: none"> <li>• Iberian lynx: Critically Endangered to Endangered after captive breeding and protection of two habitat areas (population trebled in 15 years); Fregate Island beetle after rat eradication</li> </ul>                                                                                                                                                                                                                             | 1     |
|          | <p><b>Source:</b> R. Genn, Environmental Science A level AQA (Insight &amp; Perspective, 2018), pp. 26-27; AQA spec 3.1.2.3.1.</p>                                                                                                                                                                                                                                                                                                                                         |       |

| Question | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 02.1     | <ul style="list-style-type: none"> <li>• EDGE: Evolutionarily Distinct and Globally Endangered - few close relatives, often the last species of their genus / family (Bactrian camel, pygmy hippo, northern bald ibis); loss would remove a unique branch of evolution / genes</li> <li>• Endemic: found in only one area (often islands - Galapagos, Madagascar, Seychelles: Aldabra giant tortoise, red ruffed lemur); if lost locally there is no population elsewhere to recolonise</li> <li>• Keystone: disproportionately important for the structure of the community relative to its abundance - predation of dominant species, seed dispersal, habitat creation (jaguar, forest elephants keeping paths and water holes open, wolves, beavers); losing them causes loss of many other species</li> <li>• Flagship: high public profile (tiger, panda, elephant, orang-utan) raising funds and support that also protects the whole habitat and less charismatic species (fungi, bats, moths)</li> </ul> | 8     |
| 02.2     | <ul style="list-style-type: none"> <li>• A fragmented population may consist of isolated groups each too small to be viable (resources, inbreeding) even if total numbers are large</li> <li>• A species confined to a single area (eg bonobos in one country) is vulnerable to any local change / conflict, unlike chimpanzees spread across many countries</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 28-29; AQA spec 3.1.2.3.1.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |
| 03.1     | <p><i>Any 4 from:</i></p> <ul style="list-style-type: none"> <li>• Designation of protected areas (SSSIs, MCZs)</li> <li>• Protection of wild birds and their nests (except some pests / game birds)</li> <li>• Protection of named mammals such as otters, dormice, red squirrels, badgers and their setts</li> <li>• Uprooting wild plants generally illegal</li> <li>• Bats must not be disturbed; roof timber treatments must be non-toxic to bats</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 4     |
| 03.2     | <ul style="list-style-type: none"> <li>• Appendix I: species threatened with extinction - all international (commercial) trade banned, except movement for conservation breeding</li> <li>• Examples: great apes, big cats, rhinos, blue whale</li> <li>• Appendix II: species that may become threatened if trade is not closely controlled - trade permitted under licence from countries where the species is well protected</li> <li>• Examples: Honduras mahogany, hippopotamus, green iguana, great white / basking shark, Venus fly trap</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 4     |
| 03.3     | <ul style="list-style-type: none"> <li>• Collection may be hard to police in remote areas / poor countries, but if the animals or products cannot be exported / imported there is no market</li> <li>• Removing the profit removes the incentive to collect</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2     |
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30-31; AQA spec 3.1.2.3.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |

| Question    | Marking guidance                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Marks |
|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <b>04.1</b> | <ul style="list-style-type: none"> <li>• SSSI: UK legislation (Wildlife and Countryside Act); best examples of UK flora, fauna, geological features; management plan agreed with owner who must notify Operations Likely to Damage (ploughing, drainage, pesticides, tree planting / removal)</li> <li>• SPA: EU Birds Directive; places of international importance for birds (with SACs under the Habitats Directive form the Natura 2000 network; may restrict activities in surrounding areas)</li> <li>• Ramsar: Ramsar Convention (intergovernmental agreement); internationally important wetlands, originally for waterfowl, now wetland conservation and wise use generally</li> <li>• MCZ: UK marine legislation (Marine and Coastal Access Act) under the W&amp;CA framework; protects marine habitats / species by restricting activities such as fishing and dredging; NNR: best SSSIs with complete communities</li> </ul> | 4     |
| <b>04.2</b> | <ul style="list-style-type: none"> <li>• Legal status: restrictions on damaging activities within the area (and sometimes in surrounding areas, eg drainage)</li> <li>• Management agreements between owners and statutory bodies to maintain suitable conditions</li> <li>• But it does not protect against external influences such as air pollution or climate change; very few sites are state-owned so protection depends on owner cooperation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 3     |
| <b>04.3</b> | <ul style="list-style-type: none"> <li>• Much of the UK's wildlife lives in habitats created by farming (hay meadows, hedgerows) which only survive if the farming continues</li> <li>• Managing land for wildlife costs money / reduces income so farmers need payments (points for beetle banks, hedgerows, buffer strips, wild bird seed, low-input grassland)</li> <li>• Benefits everyone so society pays through central funding; legislation cannot compel positive management</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3     |
|             | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 30, 38-39; AQA spec 3.1.2.3.2.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |
| <b>05.1</b> | • $(240\,000 - 2\,000) / 240\,000 \times 100 = 99.2\%$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | 1     |
| <b>05.2</b> | <ul style="list-style-type: none"> <li>• Whales are k-selected: slow growing, late maturing, one calf every few years, so populations recover very slowly</li> <li>• Technology (harpoon cannons, steam / factory ships) allowed catches far above the sustainable yield within decades</li> <li>• No regulation / open access to the ocean; whales were valuable (oil, spermaceti) and easily found</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 3     |
| <b>05.3</b> | <ul style="list-style-type: none"> <li>• Moratorium on commercial whaling since 1986 until populations recover</li> <li>• Total protection for certain species; whale sanctuaries</li> <li>• Limits on numbers and sizes taken; protection of suckling mothers and calves</li> <li>• Research into whale biology and threats (ship strikes, net entanglement); exceptions: aboriginal subsistence quotas (eg Inuit bowheads), 'scientific' permits (Japan - ruled not research in 2014), Norway / Iceland set their own quotas</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                | 4     |
| <b>05.4</b> | <ul style="list-style-type: none"> <li>• ITTO's 1990 target of trade only from sustainably managed forests by 2000 was not met and little progress since</li> <li>• Timber comes from many countries / private land with weak enforcement, high demand and alternative land uses; whaling involves few nations and a single treaty body</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 2     |

| Question | Marking guidance                                                                                                        | Marks |
|----------|-------------------------------------------------------------------------------------------------------------------------|-------|
|          | <b>Source:</b> R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 31-32; AQA spec 3.1.2.3.2. |       |